

2.3 Opportunities and challenges of living near coasts

Living near the coast can present people with many opportunities, like employment and leisure. However, there are also challenges associated with coastal living such as the risk of natural disasters like tropical storms and rising sea levels.

Opportunities of living near the coast

Coasts have long been associated with trade. Major ports of the world handle high numbers of imports and exports, supporting the area's economy. The Port of Jebel Ali in the United Arab Emirates, for example, was opened in 1979 and is now the busiest port in the Middle East, linking the emirate of Dubai to regional and global markets. Flat land in certain coastal areas, such as coastal plains, makes building transport infrastructure like roads and railways much easier and promotes industrial development. Fertile land on deltas and floodplains makes some areas near the sea ideal for farming. There are also fishing industries in coastal areas, with people either harvesting or processing seafood. In these ways, the coast provides many job opportunities.

Coastal areas can also be suitable locations for renewable energy projects, such as wind farms and tidal energy installations. These initiatives can encourage economic development and contribute to sustainable societies.

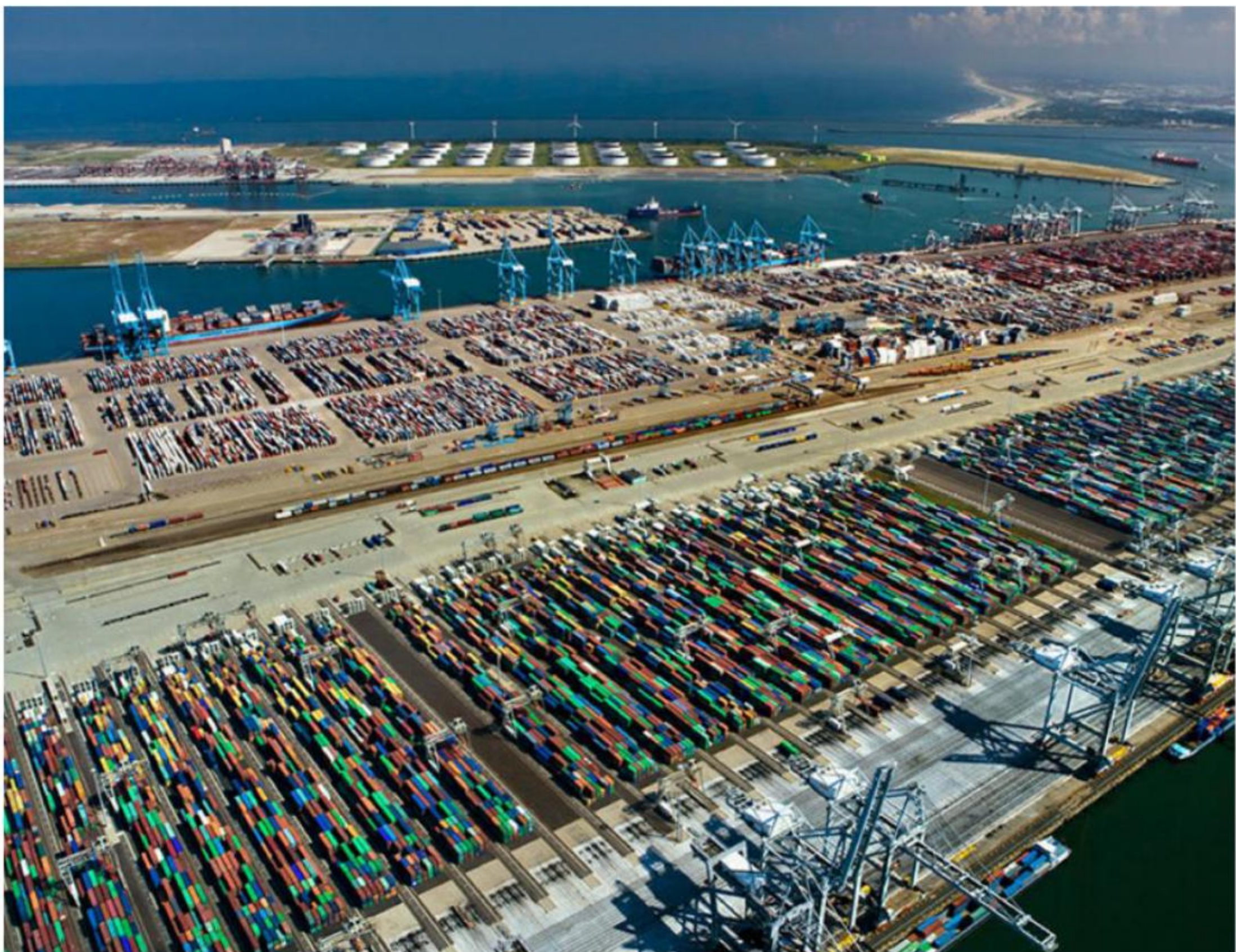


Figure 2.11: Rotterdam is Europe's largest port and handles over 14 million shipping containers each year

Coastal areas attract tourists due to their natural beauty and recreational opportunities. Living in such areas can provide easy access to beaches, water sports, and other coastal activities. Coastal nature reserves not only serve to preserve and protect coastal ecosystems and biodiversity, but they also attract visitors who want to observe wildlife. Homes near the sea are often seen as more desirable due to some of the reasons outlined above. As a result, they can have a higher value. Homes by the sea can attract affluent communities, seeking economic or recreational opportunities such as holiday rentals.

TASK 1

- a In small groups, categorise the opportunities for people living near the coast.
- b Explain why you used these categories. Is there another way to group them?

- c Within your group, discuss how these opportunities might affect each other. Consider how each of the categories you have suggested might be connected. Are these connections positive or negative?

Challenges of living near the coast

Living near the coast exposes people to various hazards, including the increased risk of natural disasters. Coastal areas are at risk from hurricanes, **storm surges**, and flooding. This can result in significant property damage and loss of life. Rising sea levels due to climate change only increases these risks. There is also the potential for **tsunamis** to occur in tectonically active coastal regions.

Coastal erosion and flooding can result in the loss of valuable land and property, threatening homes, infrastructure, and agricultural areas. Communities may be forced to relocate due to the loss of habitable land. Both flooding and erosion can disrupt coastal ecosystems, including habitats for various plant and animal species. It may lead to the loss of nesting and feeding grounds for birds, sea turtles, and other wildlife. Erosion and flooding can lead to the intrusion of saltwater into freshwater aquifers, which can affect drinking water supplies and agricultural irrigation. Coastal areas experiencing erosion are often more vulnerable to the impacts of storms and hurricanes, as protective barriers like beaches are destroyed.

THINK LIKE A GEOGRAPHER

How do the opportunities of living near coasts also present challenges for people living in coastal areas? Create a simple diagram, such as a flowchart or a feedback loop, to illustrate an example of this.

TASK 2

In pairs, debate whether the opportunities outweigh the risks of living by the coast. One person needs to argue for, and one against. After you have arrived at a conclusion as a pair, share your ideas with the rest of the class.

Sustainable management of coastal erosion and flooding

Sustainable management is necessary along coastlines to protect economically and environmentally valuable areas. These could be areas where lots of people live, areas that bring in a lot of money from tourism or fishing, or a nature reserve that is used by sea birds for breeding.

As with rivers, we can manage erosion and flooding at the coast by using hard engineering and soft engineering techniques. There are advantages and disadvantages for both. Hard engineering, although expensive, does last a long time, but usually has more of an impact on the coastline. Soft engineering is more sustainable but arguably a less effective and a more short-term approach.

Table 2.3: Hard engineering techniques to protect the coast

Technique	Description
Sea walls 	Sea walls are expensive concrete structures that protect the land from erosion and flooding. They are very effective, lasting up to 50 years, but are also expensive to maintain. Sea walls can be curved to better reflect wave energy. Some people think that sea walls are an eyesore.
Groynes 	Groynes are either made of wood or stones. They are placed at 90° to the coastline to trap sediment transported by longshore drift. This increases the size of the beach, reducing the impact of waves. Groynes can last up to 40 years. However, groynes are expensive and spoil the natural beauty of a beach. They can reduce the sediment supply to places further along the coast
Rock armour or rip-rap 	Rock armour (or rip-rap) consists of large, heavy boulders that are placed at the base of cliffs or along the coast to slow down waves and reduce wave energy. The backwash of waves drains through the gaps between the boulders, reducing sediment loss. Rock armour is costly and often has to be imported.
Revetments 	Revetments are sloped structures made of either wood or concrete and aim to reduce wave energy. Some can let water drain through them. They are inexpensive to maintain but expensive to build and spoil the landscape.
Gabions 	Gabions consist of wire cages filled with rocks. They are not as expensive as other strategies like sea walls, but they are less effective and do not last as long.
Breakwaters	A breakwater is an offshore structure that runs parallel with the shore. It aims to reduce wave energy whilst building up sediment between the beach and the breakwater. They can also provide sheltered water for ships. However, in a high tide they can be overtopped, are



expensive, and are visually unappealing. An unintentional effect is that breakwaters can also create strong currents which can impact other water users.

Table 2.4: Soft engineering techniques to protect the coast

Technique	Description
Beach nourishment 	This involves replacing material which has been lost from beaches by sea or wind erosion. The beach then acts as a natural barrier to erosion. This strategy is inexpensive but requires regular maintenance. On some beaches, this has to be done every couple of years.
Managed retreat 	Low value land is left to flood naturally. This encourages wildlife and plants, but landowners need to be compensated for their loss.

Shoreline management plans

One strategy used to manage coastlines is a **shoreline management plan (SMP)**. The approaches to SMPs vary internationally. In England and Wales, although they are not required by law, SMP documents are recommended for all sections of coastline by the Government’s Department for Environment, Food and Rural Affairs (Defra).

The four SMP policies that can be considered for each section of coastline are:

- hold the existing defence line
- advance the existing defence line
- managed realignment
- no active intervention.

Source A: Shoreline management approaches in the UK

TASK 3: Study Source A

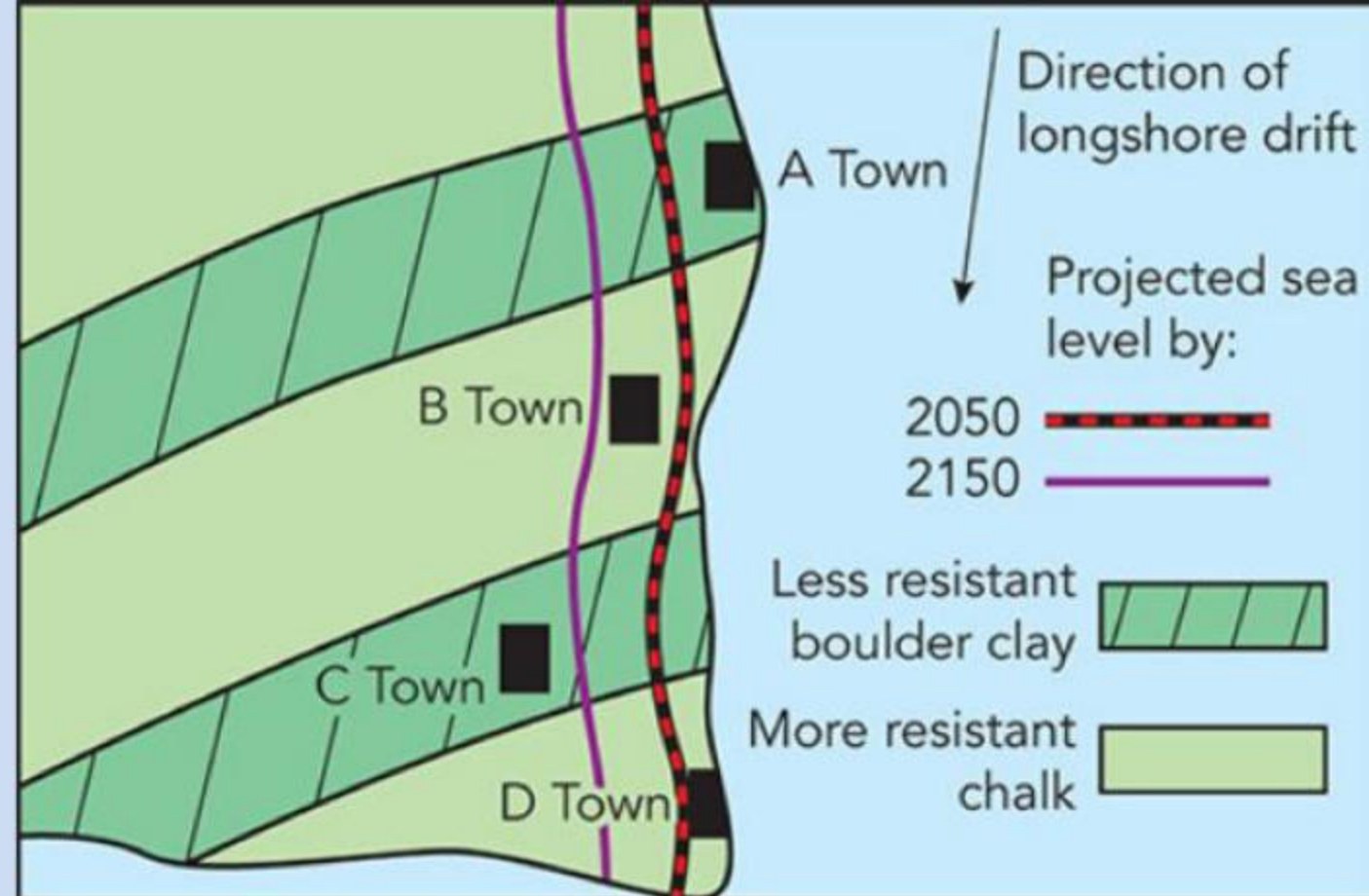
In pairs, using the extract in Source A, outline the advantages and disadvantages of the four strategic approaches of a SMP.

PROJECT

Imagine it is the year 2050 and you are in charge of creating the SMP for the imaginary coastal region in the UK shown in Source B.

In groups, firstly decide which approach you will take for each town (do nothing, managed retreat, or hold the line), then decide what you would do to manage each town. You have a budget of £30 million and your scheme needs to be in place until 2150.

Some additional information about the towns and costs is provided below. Present your ideas to the rest of the class when you have finished.



Source **B**: A map of an imaginary coastline for the project

Information about each town

Town A	A prosperous seaside town that brings in millions of pounds each year from tourism and fishing. Development here is restricted due to the threat of erosion by the sea.
Town B	Town B has always been seen as the less prosperous town to Town A. Because development is not restricted in this area, it has many local services such as a hospital and a school, with people travelling from around Town B to access these services.
Town C	A rural community living inland. The area has been in decline for some years and is in need of regeneration. The community experiences high rates of unemployment.
Town D	A busy traditional fishing port that provides jobs for locals. There is little to attract tourists here.

Costs of management techniques and their lifespan

Techniques	Overall cost (£ millions)	Lifespan (years)
Sea wall	10	50
Groynes	5	10
Rock armour	5	50
Breakwater	2	20
Revetments	2	10
Gabions	1	10
Beach nourishment	1	1
Managed retreat	1	N/A
Relocation of town residents	10	N/A

Tropical storms

Tropical storms are powerful low-pressure systems that form over warm seas and reach a wind speed of 119 km per hour. They are one of the many challenges of living in certain coastal areas. They result in very strong winds and heavy rainfall. Tropical storms are also known as hurricanes, cyclones, and typhoons, depending on where in the world they occur (see Figure 2.12). They occur between 5° and 30° north and south of the Equator, but not on the Equator, as there is no **Coriolis effect**.

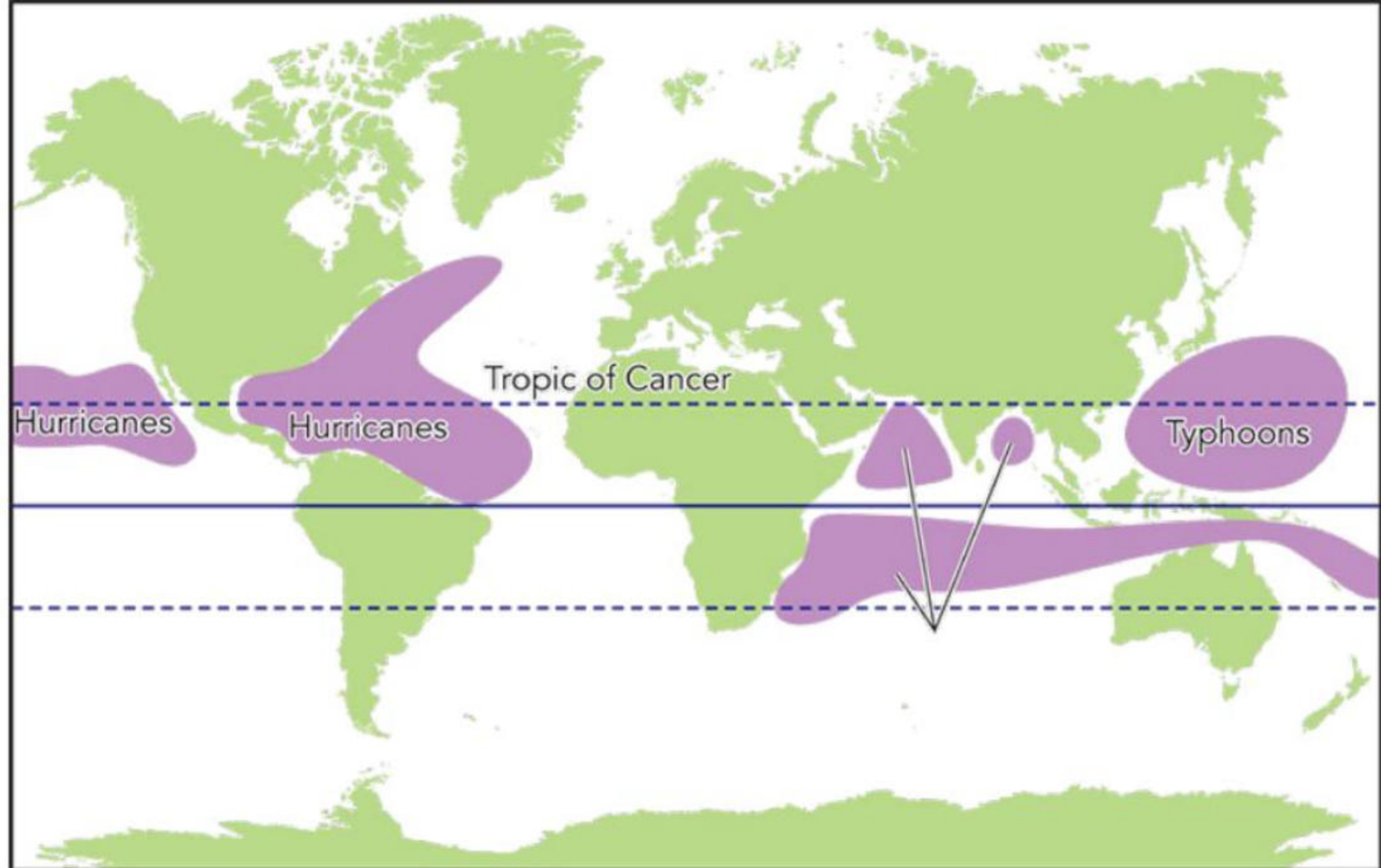


Figure 2.12: Distribution of tropical storms around the globe



Figure 2.13: Cyclone Idai caused extensive damage to coastal areas in Mozambique in 2019

The impacts of tropical storms

Tropical storms can have devastating impacts on coastal communities. The strong winds associated with these storms can destroy buildings and infrastructure. Storm surges and intense rainfall lead to coastal flooding. The subsequent rebuilding efforts and disruptions to businesses contribute to significant financial losses.

The number of deaths can also be very high, particularly in densely populated areas. Storm damages can displace people from their homes. Diseases can also spread quickly after a tropical storm if standing water becomes contaminated. In March 2019, Cyclone Idai, for example, destroyed around 715 000 hectares of crops, which increased food insecurity in Mozambique. People lost everything in the cyclone, which killed more than 1000 people and affected more than 1.5 million people in Mozambique, Malawi, and Zimbabwe. It was the second-deadliest tropical cyclone ever recorded in the Southern Hemisphere.

Managing tropical storms

Preparation, planning, protection, and prediction are strategies used to manage tropical storms and reduce their impacts.

Preparation

Preparation involves taking measures to minimise risks and ensure readiness before a tropical storm strikes. This includes developing emergency response plans,

stockpiling supplies, and educating residents about evacuation routes and safety procedures. Reinforcing buildings and securing loose objects that could become projectiles in high winds are also sensible preparatory measures.

Planning

Planning involves long-term strategies to mitigate the impacts of tropical storms. Examples include developing land-use plans to avoid building in high-risk areas and implementing building standards to ensure structures can withstand hurricane-force winds and storm surges.

Protection

Protection involves building structures to reduce the impacts of tropical storms on people and property. Building seawalls, levees, and storm surge barriers to prevent flooding are all examples of protective measures. Planting mangroves can also reduce the impacts of storm surges and coastal erosion.

Prediction

Prediction involves forecasting the path the storm will take as well as its intensity. Early warnings can then be put in place to warn vulnerable people. Meteorological agencies use satellite imagery and weather models combined with historical data to predict the development and movement of tropical storms.

Irma: Why Florida avoided catastrophe

One of the most powerful hurricanes ever recorded over the Atlantic was expected to hit Florida last week, and the state was preparing itself for some major damage. However, the devastation was mostly avoided due to the route the storm took, but also thanks to the amount of time everyone had to prepare for it. When the storm fell on the Caribbean it was the highest possible rating – category five – causing complete destruction on some islands. But by the time it reached the Florida Keys on Sunday, it was downgraded to a category four, and it continued to weaken. The state’s Governor surveyed the damage to the Keys by plane and reportedly said that he didn’t see the damage he thought he would.

Buildings in Florida are usually well-built and made to stand up to high winds; being prepared for hurricanes is a way of life. Hurricane Irma was predicted in good time, giving people a week to put up boards and storm shutters and follow the government advice to evacuate. Year on year, hurricanes batter the Caribbean; one government architect from Monserrat made a social media post to point out that, although many of the homes on the Caribbeans islands are built to withstand these storms, Irma reached speeds that were just too strong for them, and that damage from the winds and flooding was severe.

There has been criticism from island residents towards the British, French and Dutch governments for being slow to send aid to their territories. The UK’s Foreign Secretary and the French Prime Minister are due to visit the region to see the damage, and a British naval ship has arrived in the Caribbean.

Source : Some of the responses to Hurricane Irma in 2017

TASK 4: Study Source

- a Describe the short-term and long-term responses to Hurricane Irma.
- b Evaluate the different responses to Hurricane Irma.

Coral reefs

Coral reefs are often referred to as the ‘rainforests of the sea’ due to their incredible biodiversity. They are located between 30° N and 30° S in warm, salty waters that provide the ideal conditions needed for growth.

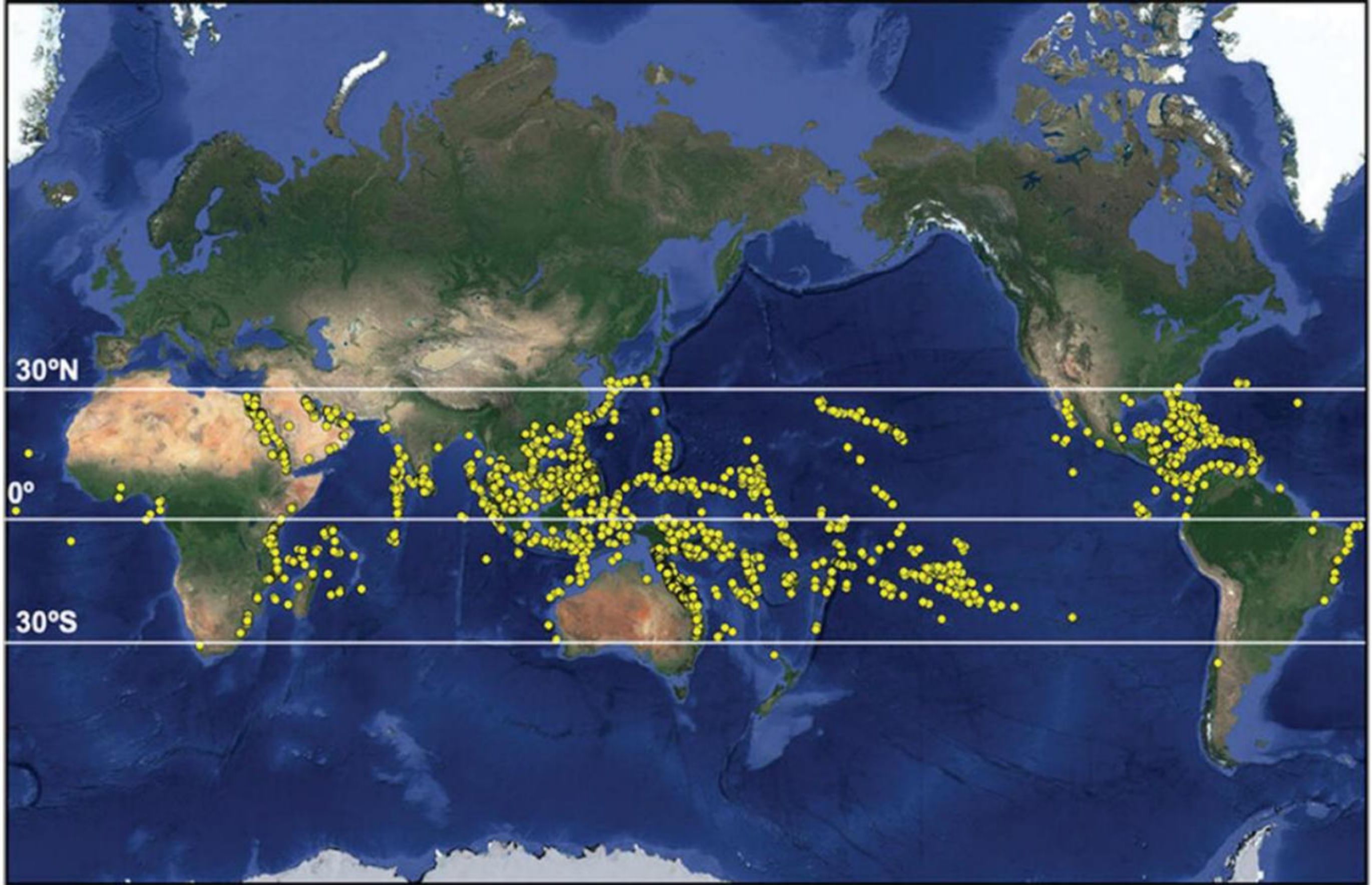


Figure 2.14: The distribution of coral reefs around the globe

The importance of coral reefs

Coral reefs are incredibly valuable, both environmentally and economically. Many commercially-important fish species rely on coral reefs for their survival. For coastal communities, coral reefs are crucial to their livelihoods and food security. Opportunities for snorkelling and scuba diving on the reef attract millions of tourists each year. Coral reefs also act as natural barriers which provide protection from storms, waves, and erosion. By helping to absorb wave energy, coral reefs reduce the impact of hurricanes and typhoons on coastal communities.

Threats to coral reefs

Many of the attractions of coral reefs can also act as threats to their sustainability. For example, swimming and diving on reefs can damage the coral. Boat anchors can be particularly destructive. Some coastal communities also trawl for fish and use explosives, which destroy the reefs. One of the biggest impacts on corals, though, is rising sea temperatures due to climate change. Increased sea temperatures can lead to **coral bleaching**. This occurs when corals expel the algae living within their tissues. These algae provide corals with essential nutrients through photosynthesis and contribute to the vibrant colours of the coral. However, when corals are stressed, they expel the algae. This leads to the loss of colour and the term 'bleaching'. If high sea temperatures persist, this can cause the coral to die.

Strategies to manage coral reefs

The management of coral reefs involves mitigating the impacts of climate change. International agreements to reduce greenhouse gas emissions can reduce the impact of coral bleaching. Some countries have introduced **Marine Protected Areas (MPAs)** to limit human activities and regulate fishing. An example of an MPA is the Galápagos Marine Reserve located around the Galápagos Islands in the Pacific Ocean. Tourist numbers are strictly limited, and visitors are required to pay an entry fee to visit the area. These funds can then support conservation projects.

In some areas, coral restoration projects have been implemented. In the Maldives Resilient Reefs project, coral propagation (growing new corals) is a technique used to restore affected reef by transplanting coral fragments to damaged areas.

Involving local communities and visitors in reef conservation efforts and providing education about the importance of coral reefs helps build awareness and support for sustainable management.



Source **D**: Scuba divers exploring a coral reef in Antigua

TASK 5: Study Source **D**

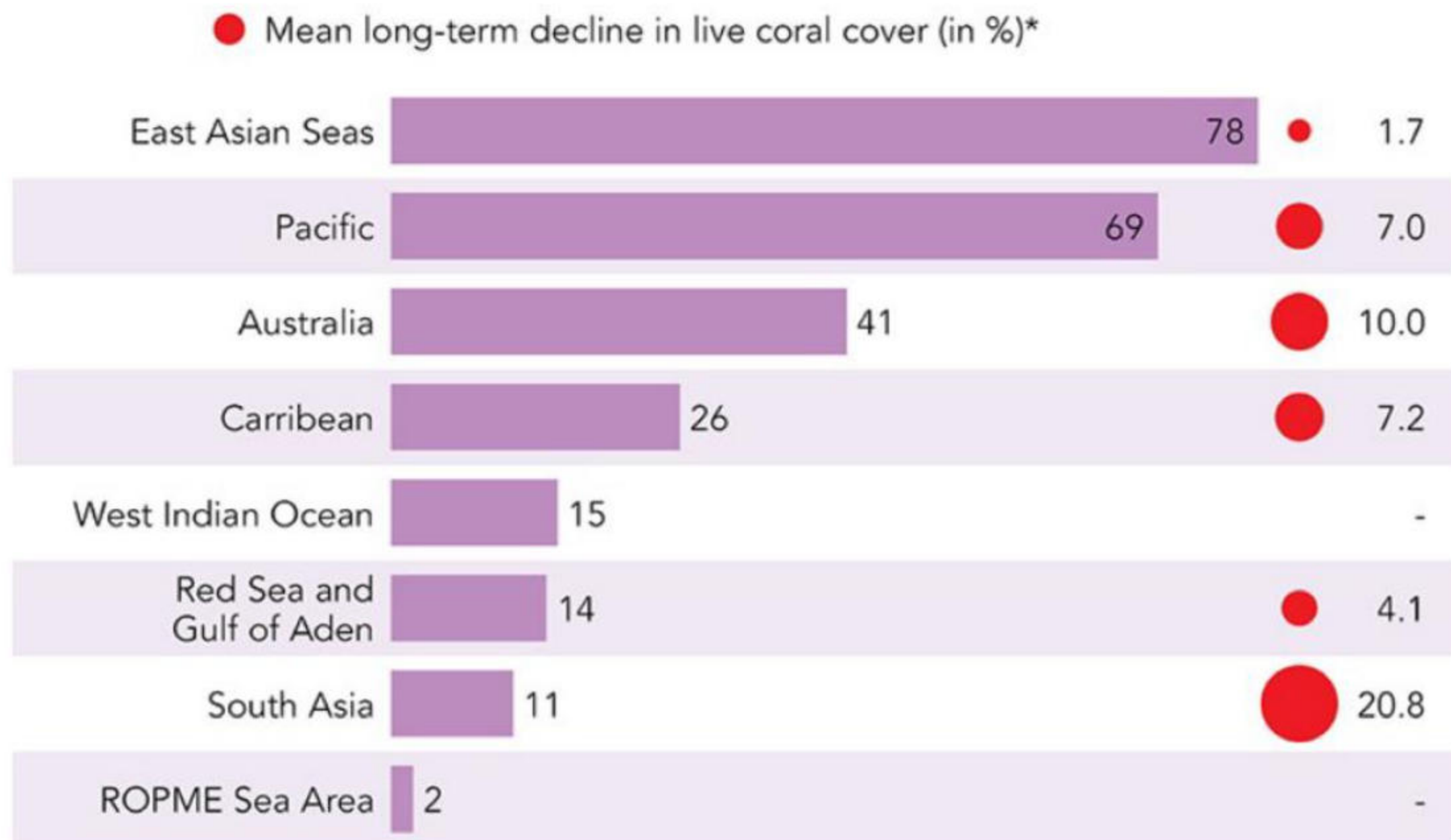
- a Outline the impacts that scuba divers could have on coral reefs.
- b Suggest how these impacts could be sustainably managed.

TASK 6: Study Source **E**

- a **Describe the distribution** and long-term decline of coral reefs globally.
- b Conduct some research to explain why South Asia has had the highest decline in living coral cover.

The Biggest Coral Reef Regions of the World

Size of largest reef areas by GCRMN region (in thousand km²)



* Difference between first survey (select regions, 1978) and most recent survey (2019).

No long-term data available for Western Indian Ocean and ROPME Sea Area.

Source: Global Coral Reef Monitoring Network

Source **E**: The biggest coral reef regions of the world

► Image Description

TIP

When describing, remember to 'say what you see'. Do not give reasons for something (for example, reasons for the global distribution of coral reefs), as this is explaining. If you are describing a graph, remember to use evidence (data) from the graph that supports what you are saying. For example, what has been the overall change? Are there any anomalies in the data?

Mangroves

Mangroves are coastal ecosystems consisting of salt-tolerant trees and shrubs that grow in brackish (slightly salty) water. Mangroves are found in tropical and subtropical coastal areas. They are another important coastal ecosystem. Their dense roots stabilise the shoreline and protect coastal communities from the impacts of tropical storms. Mangroves are also important carbon sinks that store carbon responsible for climate change. The Sundarbans mangrove forest in India, for example, stores approximately 4 000 metric tons of carbon per hectare.

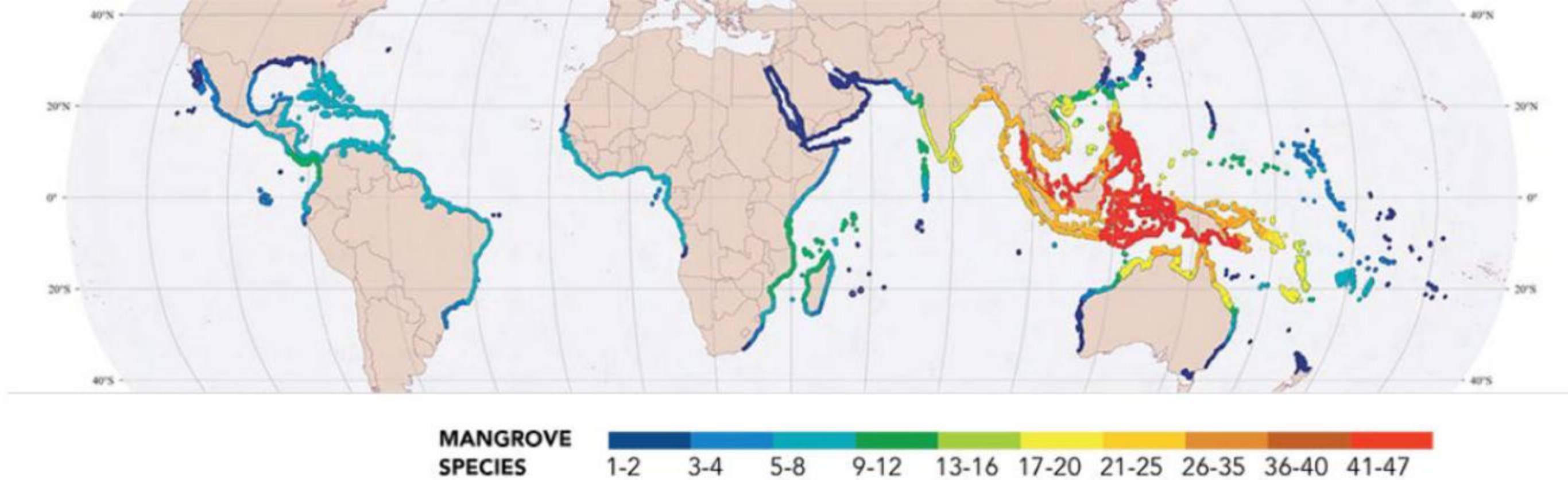


Figure 2.15: The distribution of mangroves around the globe

The threats to mangroves

Mangroves face threats such as deforestation, pollution, and climate change. Mangroves are often cleared for agricultural expansion and urban development. Countries such as Vietnam and Indonesia have experienced extensive mangrove deforestation. Pollution can also threaten mangroves. The Sundarbans mangrove forest in India faces pollution from industrial activities along the Hooghly River, impacting water quality and mangrove health. Small island nations such as the Maldives and Kiribati are particularly vulnerable to rising sea levels, which threaten their mangrove ecosystems.

Strategies to manage mangroves

One effective strategy to manage mangroves involves establishing protected areas where mangroves are conserved and protected from development and destructive activities. Promoting sustainable harvesting practices through regulations and guidelines can also be an effective management strategy.

An effective technique to manage mangroves is to involve local communities in mangrove conservation efforts through education and awareness campaigns. In Indonesia, local fishing communities have been involved in mangrove restoration projects. Community members participate in planting mangrove seedlings and patrolling protected areas to prevent illegal logging and poaching.

Through community-based initiatives, such as planting and protecting mangroves, residents can actively participate in the sustainable management of these precious ecosystems. In India, the Sundarbans Mangrove Reforestation Project aims to restore mangrove areas that have been degraded due to natural disasters and human industrial activity. Indonesia has implemented sustainable aquaculture practices such as integrated mangrove-shrimp farming systems. These systems allow for the cultivation of shrimp while preserving mangrove habitats. By adopting responsible aquaculture techniques, the negative impacts on mangroves are minimised, and the value of the coastal environment is preserved.



TASK 7: Study Source **F**

- a** Conduct some research into the adaptations of mangroves that explains how they are able to survive in the environments they are found in. Create a poster that lists at least three adaptations.
- b** **Draw a field sketch of Source F** and add it to your poster. Label the adaptations you can see in the picture that these mangroves have that enable them to survive in this environment.